Jinghang Li

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Summary

Research scientist specializing in machine learning and deep learning for large-scale, multimodal biomedical data. Experienced in developing, validating, and deploying end-to-end ML pipelines that transform research concepts into practical clinical and scientific impact.

Skills

Programming & Systems: Python, Bash, MATLAB, R

Machine Learning: Supervised and unsupervised learning, deep learning, classification, feature extraction, inference optimization, model explainbility

ML Frameworks & Libraries: PyTorch, TensorFlow, Numpy, Pandas, scikit-learn, opencv, MONAI, TorchIO

Compute & MLOps: Linux, HPC/Slurm, multi-GPU DDP, Docker, Singularity, Git

Domain Expertise: Medical imaging (3T/7T MRI), MRI reconstruction (GRAPPA)

Collaboration & Communication: Technical presentations, scientific writing, interdisciplinary collaboration, mentoring

Projects

Multi-Modal Prompt Controllable MRI Enhancement Enabling Faster Clinical Acqusition

Built an end-to-end pipeline from raw k-space to enhanced images, curating raw acquisitions, simulating physically grounded artifacts, and training flow-matching models for multi-modal image enhancement.

- Curated and organized raw Siemens TWIX (.dat) k-space across sequences; built a GRAPPA reconstruction CLI toolkit.
- Wrote custom, physically grounded k-space artifact simulation: GRAPPA undersampling (R factors), and anisotropic-resolution downsampling.
- Trained 3D prompt controllable conditional flow-matching models; enhances both fully sampled raw and post-processed (bias-corrected, denoised) images.

Autoregressive Flow Matching for Single-Step MR Image Contrast Translation (project)

Led a zero to one, clinically focused pipeline for single-step MRI contrast translation from training dataset curation to on-prem GPU deployment.

- Built a large scale paired 7T MRI dataset; scaled processing via Slurm job arrays on HPC.
- Trained conditional diffusion and flow-matching models using PyTorch/MONAI; utlized multi-GPU DDP training on HPC for faster iteration.
- Productionized inference for NIfTI/DICOM with Docker/Singularity.
- Submitted invention disclosure to the university innovation office; provisional patent filed.

niimetric — Image-Quality Metrics & MRI Artifact Simulation (project)

Open-source CLI toolkit (PyPI: niimetric) for fast quality evaluation of enhanced MRI and physics-grounded artifact simulation.

- Built a pip-installable CLI to support medical images for computing full reference metrics (PSNR, SSIM, MAE, and LPIPS)
- Added custom, physics-grounded MRI artifact simulation (coil-based noise, GRAPPA, motion, ghosting, bias field, blur)
- Published to PyPI under the MIT license for reproducible, scriptable image-quality evaluation.

Education

PhD	**University of Pittsburgh**, Bioengineering – Pittsburgh, PA, USA	Aug 2021 – present
	Carnegie Mellon University, Visiting Student, Computer Science – Pittsburgh, PA, USA	2021 – 2023
BS	**University of Pittsburgh**, Bioengineering – Pittsburgh, PA, USA	2016 – 2021

Publications

MRecover: A Conditional Generative Model for Recovering Motion-Corrupted MR Images Using AI Generated Contrast May 2026

Jinghang Li, Tales Santini, Courtney Clark, et al., Howard Aizenstein, Minjie Wu, Tamer Ibrahim

arxiv.org/abs/2605.21669 (arxiv, npj Artificial Intelligence [under review])

Bridging Neuroimaging and Neuropathology: A Comprehensive Workflow for Targeted Sampling of White Matter Lesions Oct 2025

Jinghang Li, Nadim Farhat, Jacob Berardinelli, et al., Minjie Wu, Julia Kofler, Tamer Ibrahim

https://doi.org/10.1111/jon.70094 (Neuroimaging)

A full life cycle biological clock based on routine clinical data and its impact in health and diseases Oct 2025

Kai Wang, ..., ***Jinghang Li***, ..., Xiangmei Chen, Kang Zhang

www.nature.com/articles/s41591-025-04006-w (Nature Medicine)

RF shimming strategy for an open 60-channel RF transmit 7T MRI head coil for routine use on the single transmit mode May 2025

Andrea N. Sajewski, ..., ***Jinghang Li***, ..., Tamer S. Ibrahim

10.1002/mrm.30563 (Magnetic Resonance in Medicine)

wmh_seg: Transformer based U-Net for Robust and Automatic White Matter Hyperintensity Segmentation across 1.5 T, 3T and 7T. Feb 2024

Jinghang Li, Tales Santini, Yuanzhe Huang, et al., Tamer Ibrahim, Howard Aizenstein, Minjie Wu

arxiv.org/abs/2402.12701 (arxiv)

Oral Presentation

Deep Learning–Enabled Acceleration of Submillimeter Imaging at 7T Ultra-High Field MRI 2026

Jinghang Li, Andrea Sajewski, Tales Santini, Tamer Ibrahim

Digital Health Summit, UPMC, Pittsburgh, PA

Investigating white matter hyperintensities in a multicenter COVID-19 study using 7T MRI 2023

Jinghang Li, Jr-Jiun Liou, Tales Santini, et al., Sudha Seshadri, Tamer Ibrahim

AAIC, Amsterdam, Netherlands

Patents

Image Enhancement Using Autoregressive Flow Matching Oct 2025

- U.S. Provisional Patent Application No. 63/904,914, filed October 24, 2025.

Awards

Student Venture Big Idea Competition — 3rd Place $5,000, University of Pittsburgh (April 2026)

Pitt Challenge Hackathon Winner — deep-learning facial recognition system (September 2023)

Department Best Teaching Assistant (Nominated), University of Pittsburgh (2023)